CSCS 460 A

Machine Learning

Research Paper Project

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Title: Green Parenting: Parent-Child Relations in Pakistan and the Relevant Therapeutic Models

Abstract

This research explores the psychological consequences of Authoritarian and Overprotective parenting within Pakistani families, and how children raised in these environments can identify the type of therapy most suitable for their emotional recovery. The goal of this study was to formulate a dataset and test it for the children to communicate with their parents and automatically identifying the relevant therapeutic formulations. By creating an entirely new dataset focused on Pakistani parent–child relationships, we classified emotional themes related to attachment, parenting style, maladaptive schemas, and family-system dynamics. The dataset incorporated open-ended reflections and closed-ended parenting-style ratings, which were then labelled using a multi-label therapeutic taxonomy formulated specifically for this project. Various machine-learning models were trained and evaluated, including Logistic Regression, SVM, Random Forest, MLP, and SGDClassifier. The final model was used to test real and synthetic cases to simulate how individuals might receive preliminary

therapeutic guidance. This study helps highlight the overlooked influence of Pakistani family structures, especially the Joint Family System, on mental and emotional well-being. Our findings reveal how culturally embedded forms of criticism, control, and enmeshment shape long-term psychological outcomes and demonstrate that machine-learning tools can provide useful insights into early therapeutic needs

Introduction

This paper is meant to showcase the process of our research regarding figuring out psychological therapies that people who have parents can benefit from. In Pakistan, the family dynamics are notorious for being very combative and stressful. Parents do not seem to give space to their children or in other cases they always have a standing in their child's matters. Both are very harmful forms of parenting and lead to certain traumatic imprint on children that lead them to be psychologically disturbed as they always fear the criticism since criticism is the thing that has been used on them while growing. For that reason, the problem statement of our research is " What does Authoritarian and Overprotective Parenting require for the child and how can they get counselling for further improvement?". In order to get counselling, one requires an understanding of what is truly the problem with them. Insight Therapies are based on this idea where you must uncover what the underlying issue is and that is what we plan to do in this project. Gaining the diagnosis for the therapeutic model that should be used is important for individuals to further take steps for their well-being.

This research is important as family dynamics are seen as an afterthought when it comes to the daily troubles that people have in Pakistan. This is very harmful and should not be the case as important matters need to be discussed and that is exactly what our project aims to uncover so that individuals can feel empowered to take the steps and seek the relevant therapy instead of causing

themselves psychological harm. It is also important because there is very little to no research regarding specifically the Pakistani family dynamics more specifically how Joint Family System influences children and how it innately disturbs the parents as living in a joint family means convincing their children to be a certain way that abides by their families standards, that means being extra harsh, critical and restrictive of the children.

Inputs and Outputs

Inputs (Features)

The dataset contains three types of input features: demographic information, closed-ended parenting-style ratings, and open-ended emotional reflections.

A. Demographic Features

Age Group : The participant's age range, used to understand developmental context.

Gender : The participant's gender identity.0

Family Structure : Whether the participant lives in a joint or nuclear family.

Birth Order : Whether the participant is the eldest, middle, youngest, or only child.

Marital Status : Helps understand level of independence and adult relational context.

B. Closed-Ended Parenting Style Features (Likert Scale)

These measure how the participant perceives their mother's and father's behaviours.

Mother's Parenting Items:

Strict rules (Authoritarian tendency)

Harsh punishment (Severity of discipline)

Unaware of my life (Emotional distance)

Rarely involved (Low engagement)

Allowed autonomy (Independence granted)

Warm but firm (Healthy authoritative parenting)

Father's Parenting Items:

Same themes as mother, but evaluated separately.

Additional Closed-Ended Items:

Academic/career pressure from mother

Academic/career pressure from father

Teenage stress level

Communication quality with mother

Communication quality with father

Overall emotional support from parents

Relationship satisfaction with parents

C. Open-Ended Narrative Features

These are the main text inputs used by the machine-learning models.

Valued Reflection : A short description of a positive or meaningful experience with parents.

Conflict Reflection : A description of a challenge, conflict, or emotional difficulty with parents.

These narrative responses contain emotional and relational cues that help the model infer psychological patterns.

Output (Target Variable)

The target variable is a multi-label therapeutic classification called Therapeutic Focus Labels.

Each participant may have more than one therapeutic label.

The label space includes 23 therapeutic constructs, grouped into psychological domains.

Attachment Patterns:

Anxious Attachment

Avoidant Attachment

Disorganized Attachment

Attachment Injury / Rupture

Parenting Style Patterns:

Authoritarian Parenting

Permissive / Inconsistent Parenting

Overprotective / Enmeshing Parenting

Detached / Emotionally Unavailable Parenting

Schema Patterns:

Emotional Deprivation Schema

Subjugation Schema

Unrelenting Standards Schema

Failure Schema

Family System Patterns:

Parentification

Triangulation

Boundary Diffusion

Intergenerational Trauma Pattern

Emotional Processing:

Suppressed Emotional Expression

Conflict Avoidance Conditioning

Internalized Criticism and Shame

Therapeutic Needs:

Needs Emotion Coaching

Needs Boundary Re-Education

Needs Healthy Autonomy Development

Needs Communication Skills Repair

Literature Review

Roe and Siegelman created one of the earliest tools for examining the parent–child relationship, focusing on warmth, control, and emotional attitudes between parents and children (Roe & Siegelman, 1963). Their work highlighted that children internalise parental behaviours as emotional schemas that influence long-term personality and functioning. While foundational, their scale was quantitative, Western in nature, and did not address culturally specific forms of authoritarianism or enmeshment found in South Asian households. Our study

differs by capturing open-ended reflections, allowing for a richer emotional expression, and by building a therapeutic label space reflecting Pakistani cultural and relational system.

Hair and colleagues developed a validated scale to understand warmth, communication, and conflict within parent–adolescent interactions (Hair et al., 2008). The scale captures important relational dimensions but remains confined to structured questions rather than narrative-style responses. Their work served as groundwork for identifying emotional closeness and autonomy needs. However, unlike our project, their scale does not classify responses into therapeutic categories nor does it use machine learning. Our research builds upon their relational domains but provides a more applied therapeutic angle through automated label detection using open narrative data.

Dixson et al. Introduced the Parent–Child Relationship Schema Scale, examining expectations and internal schemas formed through parent–child interactions (Dixson et al., 2014). Their work aligns closely with our project's theoretical foundation because it recognises schemas as internalised patterns shaping emotional life. However, their study still used structured items, whereas ours processes unstructured text narratives and classifies them using multi-label psychological formulation. Their work inspired the schema components of our label taxonomy but does not extend to machine-learning-based therapeutic recommendations.

Antonucci et al. Demonstrated that machine learning could classify psychosis risk using attachment and parenting variables (Antonucci et al., 2021). Their study proves that parenting features can be used for predictive modelling in mental health. However, their dataset consisted of clinical populations and quantitative assessment tools, whereas our dataset is culturally grounded and includes emotionally expressive reflections. Their findings support that psychological constructs like attachment are ML-predictable, validating the

direction of our research, but our project focuses on therapy identification, not clinical diagnosis.

Sun and Zhao used machine learning to link parental behaviours with academic performance (Sun & Zhao, 2024). Their work shows ML can detect psychological influences on children through structured parental variables. While valuable, their approach remains academic-performance oriented rather than therapeutically oriented. Our project expands the scope by mapping emotional narratives to therapeutic frameworks, filling a gap in culturally specific parental-relationship ML research.

Xiaoran Sun discusses how supervised ML can support exploratory family research by discovering underlying relational patterns (Sun, 2024). This directly supports the methodological foundation of our project. However, unlike our dataset, their work relies on existing validated scales, whereas we constructed a novel dataset with richer emotional content. Additionally, Sun's work does not involve therapeutic labelling. Our project stands apart by bridging ML and psychotherapy.

The NURTURE model used ML to evaluate factors influencing healthy parent–child relationships and provided recommendations (Saraswat et al., 2024). Their study is closest to ours conceptually but is still different in important ways. Their model is predictive and prescriptive, while ours is diagnostic and therapeutic, focusing on identifying internal emotional wounds and schema patterns rather than general relational health. Their dataset is also not culturally specific. Our work extends ML-based family analysis into a psychotherapeutically meaningful and Pakistan-specific direction.

Importance and Significance of Work Done

Across the reviewed literature, it becomes clear that the parent–child relationship plays a fundamental role in shaping emotional development,

attachment styles, and long-term psychological functioning. Classic tools such as the Parent–Child Relations Questionnaire (Roe & Siegelman, 1963) and contemporary scales like the Parent–Adolescent Relationship Scale (Hair et al., 2008) have demonstrated that parental behaviours directly influence children's internal schemas, communication patterns, and emotional well-being. More recent studies using machine learning, such as Antonucci et al. (2021) and Sun & Zhao (2024), further show that parenting variables can be used to predict psychological outcomes or academic consequences. Together, these works highlight the importance of systematically studying parenting environments to understand psychological risks and relational outcomes. However, nearly all existing research—and especially all machine-learning applications—are based on Western or non-Pakistani contexts, making our work significant because it introduces culturally relevant parent–child relationship data and applies machine learning to generate therapeutic insights grounded in Pakistani family dynamics. This contributes not only to academic understanding but also to the practical field of culturally aligned psychological support.

Why Our Solution is Needed

While previous research provides valuable frameworks for measuring parent–child relationships, none of the reviewed studies address the specific emotional realities of Pakistani households or the therapeutic implications of authoritarianism, criticism, overprotectiveness, joint-family pressures, and emotional suppression. Existing tools rely heavily on structured questionnaires and do not capture the nuanced emotional language that individuals use when describing meaningful or painful experiences with their parents. Furthermore, no available research translates parent–child relational patterns into therapeutic formulations, leaving individuals without clear guidance on what type of counselling or intervention might suit their emotional needs. Our solution fills this gap by using machine learning to analyse natural open-ended reflections

and automatically identify therapeutic constructs such as attachment wounds, maladaptive schemas, boundary violations, and unmet emotional needs. In a society where mental health is neglected, misunderstood, or stigmatized, and where family dynamics profoundly shape psychological well-being, our approach offers an accessible, culturally relevant, and actionable tool to help individuals understand what kind of therapy they may need and why. This makes our project both necessary and transformative in the context of Pakistan's mental-health landscape.

Plan

Our plan is to conduct a google forms survey and gather data using the Features described above and using their Open ended description to infer the kind of psychological therapy or model they should follow. This model would be annotated to our data by a Psychology major student who will annotate the instances of our dataset based on his Psychology centered knowledge.

Methodology

Dataset Construction:

We designed a Google Form collecting:

(1) demographic data,

(2) closed-ended parenting-style items, and

(3) two open-ended reflections describing valued and conflict-based parental experiences.

The dataset contained 83 real responses.

Therapeutic Label Space:

Using Attachment Theory, Schema Therapy, Parenting Style typologies, Family Systems Theory, Emotional Processing Theory, and Therapeutic Needs

literature, we created a hierarchical multi-label taxonomy of 23 therapeutic constructs.

Annotation Process:

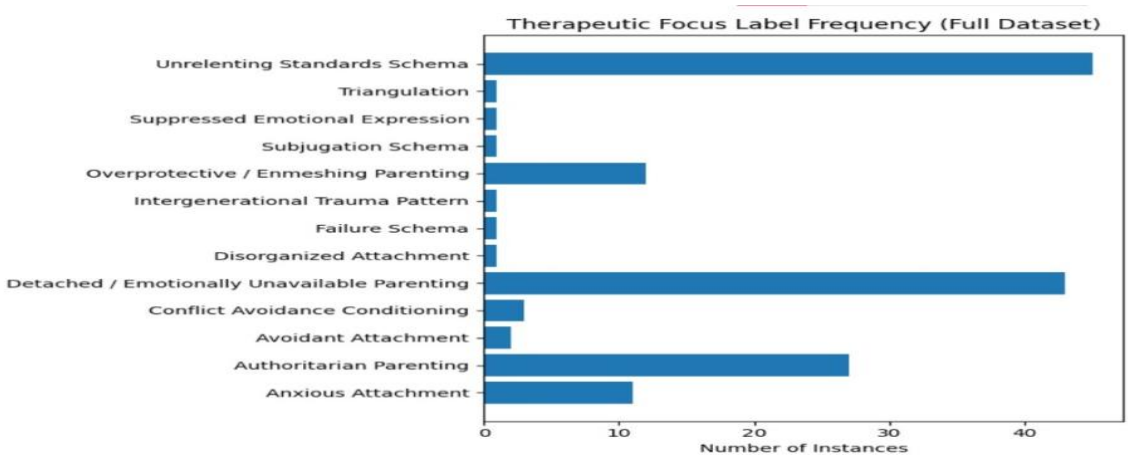
We were able to get all 83 instances of our dataset manually labelled by an undergraduate Psychology major student.

Machine Learning Models

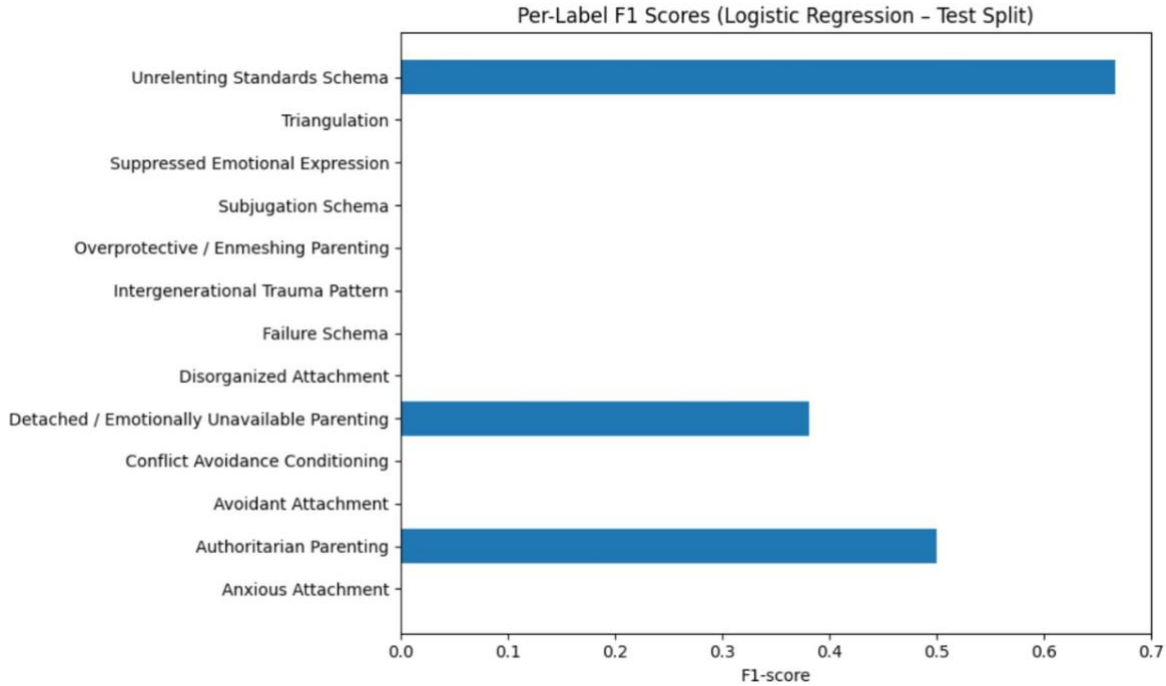
We narrowed down our testing space to these models as they seemed most appropriate:

Model	**Reason**
Logistic Regression	Strong baseline for high-dimesional text, excellent interpretability
Linear SVM	Works extremely well with sparse text features, baseline comparison
Random Forest	Non-linear model capturing complex relational patterns
MLPClassifier	Neural baseline capturing non-linearities in TF-IDF space
SGDClassifier	Efficient linear model, alternative optimizer to LogReg

Experiments/Results



This plot shows that the dataset is heavily imbalanced, with Unrelenting Standards Schema, Detached/Emotionally Unavailable Parenting, and Authoritarian Parenting appearing far more frequently than other labels. Several therapeutic categories, such as Triangulation, Subjugation Schema, and Suppressed Emotional Expression, occur only rarely. This imbalance strongly influences downstream model performance.



The model achieves meaningful F1-scores only for high-frequency labels, particularly Unrelenting Standards Schema, Detached/Emotionally Unavailable Parenting, and Authoritarian Parenting. All low-support labels have an F1-score of zero, indicating that the model cannot generalise on categories with insufficient training examples.

```
=== Linear SVM – Classification Report (Full Dataset Split) ===
                                        precision    recall  f1-score   support

...
                          Anxious Attachment       0.00      0.00      0.00         3
                      Authoritarian Parenting       1.00      0.14      0.25         7
                          Avoidant Attachment       0.00      0.00      0.00         0
                 Conflict Avoidance Conditioning       0.00      0.00      0.00         2
Detached / Emotionally Unavailable Parenting       0.44      0.33      0.38        12
                      Disorganized Attachment       0.00      0.00      0.00         0
                               Failure Schema       0.00      0.00      0.00         1
             Intergenerational Trauma Pattern       0.00      0.00      0.00         0
          Overprotective / Enmeshing Parenting       0.00      0.00      0.00         3
                         Subjugation Schema       0.00      0.00      0.00         0
              Suppressed Emotional Expression       0.00      0.00      0.00         0
                               Triangulation       0.00      0.00      0.00         0
                   Unrelenting Standards Schema       0.53      0.90      0.67        10

                                   micro avg       0.52      0.37      0.43        38
                                   macro avg       0.15      0.11      0.10        38
                                weighted avg       0.46      0.37      0.34        38
                                 samples avg       0.40      0.31      0.32        38
```

The SVM demonstrates high performance only on Unrelenting Standards Schema, while most other labels have precision and recall of zero. This reflects the model's sensitivity to class imbalance and highlights its difficulty in learning from sparse textual data when many classes appear infrequently.

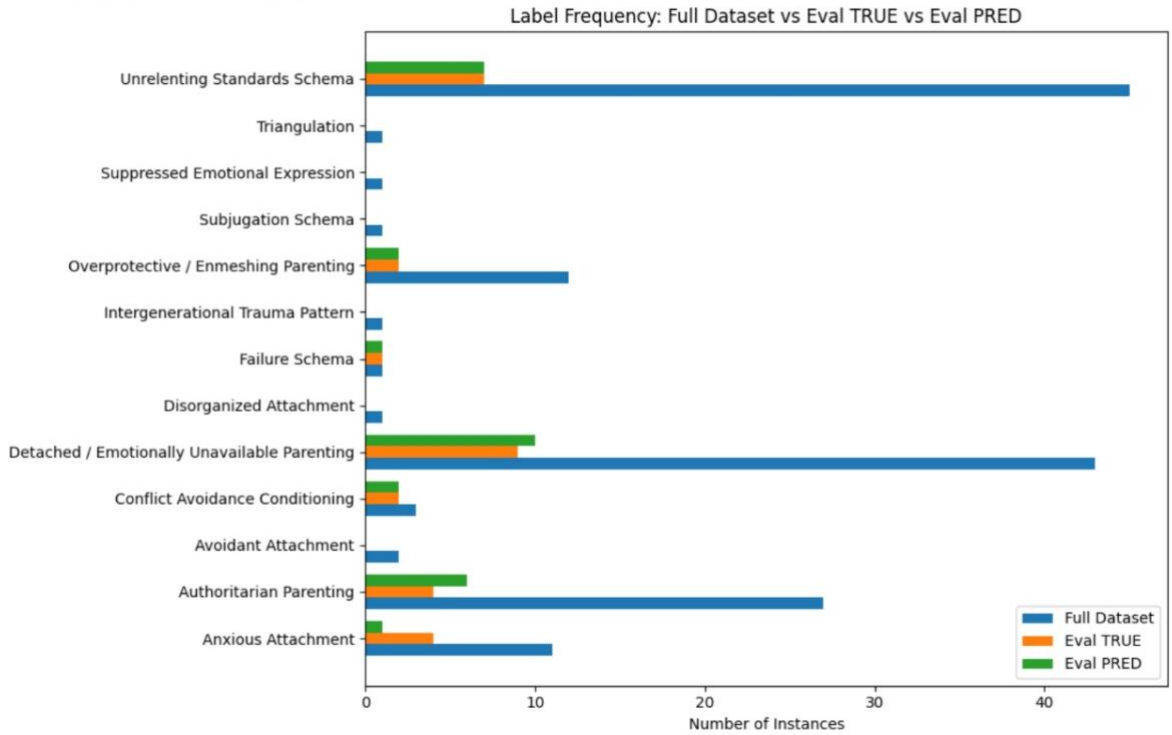
```
=== Evaluation Set - Classification Report (20 Instances) ===
                                          precision    recall  f1-score   support

                       Anxious Attachment       1.00      0.25      0.40         4
                 Authoritarian Parenting       0.67      1.00      0.80         4
                     Avoidant Attachment       0.00      0.00      0.00         0
          Conflict Avoidance Conditioning      1.00      1.00      1.00         2
Detached / Emotionally Unavailable Parenting   0.80      0.89      0.84         9
                 Disorganized Attachment       0.00      0.00      0.00         0
                          Failure Schema       1.00      1.00      1.00         1
         Intergenerational Trauma Pattern      0.00      0.00      0.00         0
     Overprotective / Enmeshing Parenting      0.50      0.50      0.50         2
                     Subjugation Schema        0.00      0.00      0.00         0
          Suppressed Emotional Expression      0.00      0.00      0.00         0
                         Triangulation        0.00      0.00      0.00         0
             Unrelenting Standards Schema     0.86      0.86      0.86         7
```

Here is the evaluation of the dataset trained model on set of 20 instances from within the dataset but unmarked with their label.

```
Eval PREDICTED label counts:
 Anxious Attachment                                  1
Authoritarian Parenting                              6
Avoidant Attachment                                  0
Conflict Avoidance Conditioning                      2
Detached / Emotionally Unavailable Parenting        10
Disorganized Attachment                              0
Failure Schema                                       1
Intergenerational Trauma Pattern                     0
Overprotective / Enmeshing Parenting                 2
Subjugation Schema                                   0
Suppressed Emotional Expression                      0
Triangulation                                        0
Unrelenting Standards Schema                         7
dtype: int64

Cosine similarity (Full vs Eval TRUE): 0.968
Cosine similarity (Full vs Eval PRED): 0.972
```

Predicted label counts closely match the true label counts for the evaluation set. High cosine similarity values (0.968 for true vs. full dataset, 0.972 for predicted vs. full dataset) indicate that both the evaluation set and the model's predictions preserve the overall therapeutic structure of the full dataset

```
Cosine similarity (Full vs Eval PRED): 0.972
```



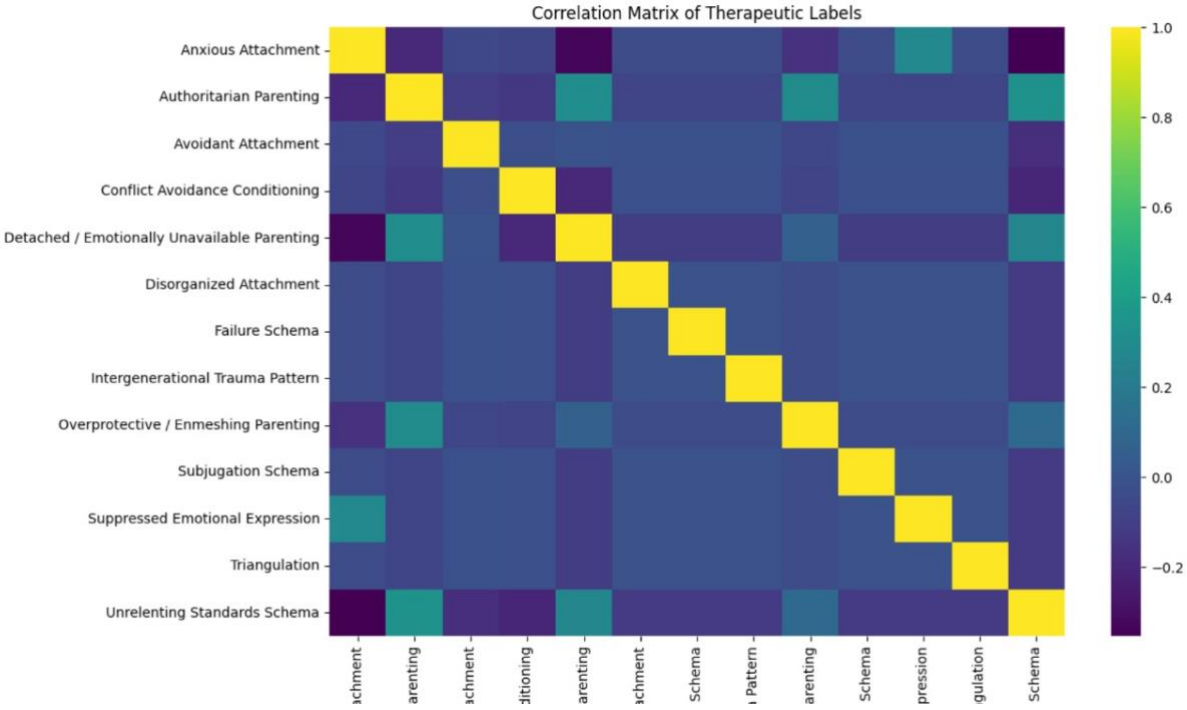
The three-way comparison shows strong alignment between the full dataset and evaluation subset, and between evaluation true labels and model predictions. High-frequency labels dominate across all three views, demonstrating that the model maintains distributional consistency with the dataset.

```
=== Logistic Regression - Classification Report (Full Dataset Split) ===
                                          precision    recall  f1-score   support

                       Anxious Attachment       0.00      0.00      0.00         3
                  Authoritarian Parenting       0.60      0.43      0.50         7
                      Avoidant Attachment       0.00      0.00      0.00         0
           Conflict Avoidance Conditioning       0.00      0.00      0.00         2
Detached / Emotionally Unavailable Parenting       0.44      0.33      0.38        12
                  Disorganized Attachment       0.00      0.00      0.00         0
                            Failure Schema       0.00      0.00      0.00         1
         Intergenerational Trauma Pattern       0.00      0.00      0.00         0
      Overprotective / Enmeshing Parenting       0.00      0.00      0.00         3
                     Subjugation Schema       0.00      0.00      0.00         0
          Suppressed Emotional Expression       0.00      0.00      0.00         0
                           Triangulation       0.00      0.00      0.00         0
             Unrelenting Standards Schema       0.53      0.90      0.67        10

                               micro avg       0.50      0.42      0.46        38
                               macro avg       0.12      0.13      0.12        38
                            weighted avg       0.39      0.42      0.39        38
                             samples avg       0.41      0.36      0.35        38
```

Logistic Regression continues to perform well on high-frequency labels and poorly on rare labels. The micro-average scores are higher than macro-average scores, indicating that performance is driven by the most common therapeutic categories rather than balanced across all labels.

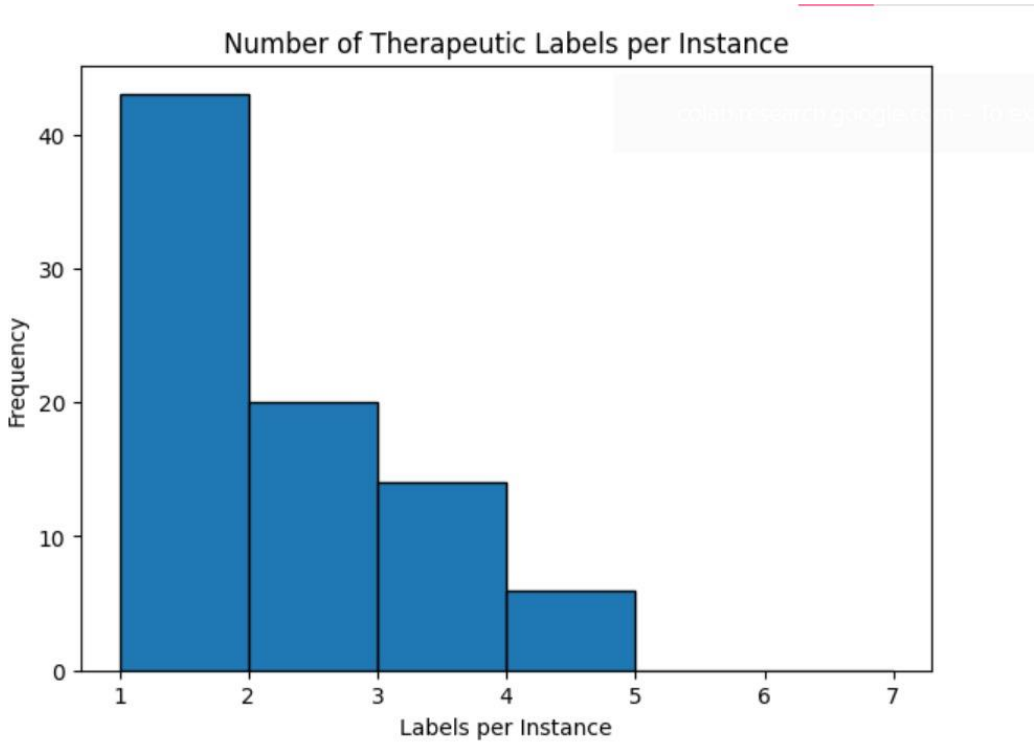
```
=== MLPClassifier - Classification Report (Test Split) ===
                                          precision    recall  f1-score   support

                       Anxious Attachment       0.00      0.00      0.00         3
                  Authoritarian Parenting       1.00      0.14      0.25         7
                      Avoidant Attachment       0.00      0.00      0.00         0
            Conflict Avoidance Conditioning       0.00      0.00      0.00         2
Detached / Emotionally Unavailable Parenting       0.29      0.17      0.21        12
                  Disorganized Attachment       0.00      0.00      0.00         0
                           Failure Schema       0.00      0.00      0.00         1
           Intergenerational Trauma Pattern       0.00      0.00      0.00         0
       Overprotective / Enmeshing Parenting       0.00      0.00      0.00         3
                     Subjugation Schema       0.00      0.00      0.00         0
          Suppressed Emotional Expression       0.00      0.00      0.00         0
                         Triangulation       0.00      0.00      0.00         0
              Unrelenting Standards Schema       0.47      0.70      0.56        10

                                 micro avg       0.43      0.26      0.33        38
                                 macro avg       0.13      0.08      0.08        38
                              weighted avg       0.40      0.26      0.26        38
                               samples avg       0.36      0.24      0.25        38
```

The neural model struggles significantly, achieving meaningful performance only on Unrelenting Standards Schema. Low recall on most labels reflects overfitting risks and the limitations of training neural networks on a small, sparse dataset with many low-support classes.



The heatmap shows that several therapeutic labels frequently appear together, forming meaningful psychological clusters. Detached/Emotionally Unavailable Parenting, Authoritarian Parenting, and Unrelenting Standards Schema display high co-occurrence, indicating that children who experience emotional

disconnection from parents often also describe strictness and pressure. Conversely, many label- such as Triangulation, Subjugation Schema, and Suppressed Emotional Expression-show minimal co-occurrence with other categories, revealing their rarity and low representation within the dataset. These patterns suggest that the dataset is structured around a core set of interlinked relational wounds.



Most participants exhibit one or two major therapeutic concerns, while fewer instances display three or more labels. This distribution shows that although the dataset is multi-label, the majority of responses centre around a primary emotional theme. The long-tail distribution reflects natural psychological variation but also reinforces that many therapeutic constructs are underrepresented during training.

Across all models, performance remains modest due to dataset imbalance, but SGD and Logistic Regression outperform the others, followed by SVM, Random Forest, and MLP. The comparatively higher F1-scores achieved by SGD and Logistic Regression reflect their suitability for high-dimensional sparse text data. Neural and non-linear models perform worse, likely because

they require larger datasets to capture subtle multi-label patterns. The graph highlights that simpler linear models remain the most effective approach for this task.

Discussion

The findings of this study reveal several important insights into both the therapeutic landscape of Pakistani parent-child relationships and the feasibility of applying machine-learning techniques to psychologically meaningful multi-label classification. The dominance of a small group of therapeutic categories (particularly Unrelenting Standards Schema, Detached/Emotionally Unavailable Parenting, and Authoritarian Parenting) reflects recurring emotional and relational themes within the dataset. These patterns align with culturally recognised dynamics in Pakistani households, such as high academic pressure, emotional restraint, and strict behavioural expectations. The strong co-occurrence among these labels further indicates that these psychological experiences often emerge together, forming a relational profile characterised by emotional distance, heightened criticism, and unmet childhood needs. The dataset therefore captures a coherent therapeutic structure that mirrors real-world familial relational patterns.

However, the significant class imbalance across therapeutic constructs had a notable impact on model behaviour. Machine-learning algorithms performed reliably on high-frequency labels and consistently failed to identify low-support labels such as Subjugation Schema, Triangulation, and Intergenerational Trauma Pattern. This outcome reflects the challenges of learning complex, nuanced emotional constructs from limited examples. Linear models (namely Logistic Regression and SGDClassifier) outperformed non-linear and neural models because they are better suited to sparse, high-dimensional text data and can extract clearer signal from a small dataset. Despite modest overall macro F1-scores, the very high cosine similarity between full-dataset label distributions

and both evaluation true labels and predicted labels suggests that the model internalised the broad therapeutic structure effectively. It recognised central emotional patterns even when finer-grained schemas remained difficult to classify. Collectively, the results show that the proposed approach is promising for identifying dominant therapeutic themes but requires further data expansion to fully capture the psychological diversity of parent-child relational experiences.

Conclusion

This research demonstrates the potential for machine-learning models to assist in identifying therapeutic constructs within narratives describing Pakistani parenting experiences. By developing a culturally grounded dataset and designing a multi-label therapeutic taxonomy, the research provides a foundation for automated detection of psychological themes such as emotional unavailability, authoritarianism, and internalised pressure. The results indicate that the models effectively captured the dominant therapeutic patterns embedded in the dataset, as shown by high distributional similarity and consistent performance on frequently occurring labels. This suggests that machine-learning tools can play a meaningful role in helping individuals gain clarity about their emotional experiences and possible therapeutic directions, especially in settings where mental-health awareness is limited.

At the same time, the study highlights key limitations and areas for future work. The dataset's imbalance and small size restricted the models' ability to generalise to rare therapeutic constructs, indicating a need for larger and more diversified data collection to capture the full range of psychological experiences. Rare patterns (such as systemic family-role distortions and deep-seated schemas) require more examples for models to learn reliably. Future research may benefit from incorporating richer linguistic features, transformer-based models, and expanded participant diversity to strengthen generalisability.

Overall, the project establishes an important step toward culturally relevant, technology-assisted psychological insight generation and demonstrates the feasibility of combining machine learning with therapeutic formulation frameworks tailored to Pakistani family dynamics

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